energywave.ai • Thread C — measurement platform error budget, as a template for commissioning
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CONFIDENTIAL — DRAFT — FOR DISCUSSION · BOUNDS ARE MEASURED AT COMMISSIONING, NOT MODELLED HERE

The platform is the asset whatever the gates find. Its error budget is a table of measured upper bounds.

The September prospectus commits the company to publish the measurement platform's design and its error budget as a methods paper before any result is claimed. The budget assigns to every known way of manufacturing a picowatt-level signal a *measured* upper bound on the spurious output it could produce in the platform as built. The artifact floor is the conservative sum of those bounds, with a separately reported combined standard uncertainty for the statistical part; the platform is not commissioned until the floor is at least a factor of ten below the effect it is built to test. This page is that table with the method for each row and the bounds left to be measured.

1 · THE COMMISSIONING CRITERION

Measurand. Electrical output power delivered to a defined load under stated operating conditions (bias, temperature, shielding, load impedance). Every row of §2 is expressed in that measurand: a bound on a current or a voltage is converted to power through the load model before it enters the budget. **Artifact floor** = Σ $|\text{bound}_i|$ over the numerical rows of §2 — the conservative arithmetic sum of absolute bounds, because systematic artifacts can add with the same sign; root-sum-square is used only for components shown to be independent, and the combined standard uncertainty u_c (standard uncertainties with stated distributions, coverage and covariance, per the NIST guidance) is reported beside the floor, never in place of it. Procedural rows are controls and are excluded from the arithmetic. Each bound is measured on dead devices, twins and fixture-only runs in the platform as built. **Commissioned** when floor ≤ (reported effect) ÷ 10 — the September prospectus §6.3 rule, inherited and tightened here. For thread B the reported effect is 1.4 pW (Moddel et al. 2021 — inferred load power, one laboratory, not independently replicated in the reviewed evidence), so the plan proposes a floor at or below 0.14 pW — a commissioning target set by this plan, not an established requirement; the thread C lead and the advisory board ratify it at commissioning. For thread A the effect is the smallest conditioning-dependent change the pre-registration declares meaningful. **[Needs Input from Ian or Chip: the thread A effect size the plan must resolve]**

2 · THE ERROR-BUDGET TABLE

Artifact source	How the bound is measured	Instrument / fixture	Measured bound
Thermal gradient (Seebeck)	Map ΔT across the device site with distributed sensing; sweep ±ΔT deliberately; record output of dead devices and no-cavity twins versus ΔT; bound = output at the residual gradient after stabilization.	Multi-stage actively stabilized isothermal enclosure; distributed temperature sensing	[to be measured] W at residual ΔT = [ ] mK
Photo-effects	Measure light leakage in the closed enclosure; sweep controlled illumination; bound = output at measured leakage on mirror-only twins.	Light-tight enclosure; calibrated photodiode; controlled source	[to be measured]

Artifact source	How the bound is measured	Instrument / fixture	Measured bound
Electromagnetic pickup and rectification	Inject known RF at swept frequency and power; record device response; measure residual ambient field inside shielding; bound = response at residual ambient.	Nested conductive and high-permeability shielding; filtered feedthroughs; guarded triaxial wiring; spectrum monitor	[to be measured]
Contact potential and electrochemistry	Humidity-controlled and vacuum runs; discharge-and-recovery tests; long-term drift on twins; bound = steady residual after discharge.	Humidity control; vacuum option; electrometer	[to be measured]
Stored charge and dielectric relaxation	Extended discharge to ground before every run; relaxation-time fits on spacer-only twins; bound = residual current after the declared discharge interval.	Electrometer; timed grounding fixture	[to be measured] (a current — converted to power through the load model)
Triboelectric and piezoelectric	Correlate output with accelerometer spectrum; isolation on and off; bound = output at the residual vibration spectrum.	Passive isolation; accelerometers	[to be measured]
Instrument offsets and bias current	Interleave dead-device, open-circuit and shorted runs with every measurement; characterize offset drift over the run length; bound = drift-corrected residual.	Electrometer-class instrumentation with characterized bias current	[to be measured]
Leakage paths	Fixture-only runs; guarded fixtures; bound = fixture output with no device.	Guarded fixture	[to be measured]
Probe phase skew (power measurements)	Full-chain calibration; probe swap; frequency sweep against the deskew model; bound = residual real-power error at the highest frequency used.	Sampled v(t), i(t) acquisition; deskew fixture	[to be measured]
Selection and analysis bias	Not a measured bound: pre-registered device matrix, blinding keys held outside the team, fixed stopping rules, all devices reported.	Registry deposit; data pipeline with hashing at acquisition	procedural
Artifact floor	**Σ |bound$_i$|** over the numerical rows = [to be computed at commissioning], with u_c reported beside it — commissioned when ≤ 0.14 pW for thread B (proposed target).		

3 · COMMISSIONING SEQUENCE

1. Build and close the enclosure; measure shielding effectiveness versus frequency and the residual ambient field.
2. Map the thermal field at the device site over 24 h; set the stability and residual-gradient targets in the pre-registration.
3. Characterize the electrometer chain on open, short and known-null loads; record bias current and offset drift.
4. Run dead devices, fixture-only and each twin class through the full measurement sequence; fill every bound in §2 from those runs.

5. Compute the floor; compare with the criterion; publish the methods paper with the table; only then measure a live device.

4 · CALORIMETRY — COMMISSIONED IN GATE 1, DECISIVE IN GATE 2

A differential calorimeter with a reference cell alongside the device cell, so that heat flow into the device can be measured while output is drawn. Its resolution sets what Gate 2 can say: if output is too small to resolve calorimetrically, Gate 2 requires the device count and integration time needed to resolve it before any conclusion is drawn. Resolution target **[Needs Input from Ian or Chip: the calorimetric resolution the program will fund]** .

5 · SIGN-OFF

Role	Responsibility	Named
Measurement and instrumentation lead (thread C)	Owns the table, the commissioning runs and the methods paper.	**[Needs Input from Ian or Chip: name, or “to hire — a Tranche A condition”]**
Principal investigator	Signs the pre-registration; certifies the floor before live devices.	**[Needs Input from Ian or Chip: name, or “to hire”]**
External scientific advisory board	Reviews the table before publication; holds the veto over performance statements.	To recruit — at least one declared critic

Sources: Technical Prospectus v1 §§6.3–6.4, 7.1, Appendix B (M-05); Physical Architecture & Baseline Model v1.3 §3 (M-14); Investor Overview slides 10, 12 (M-04); Research Program and Gates page; Thread A plan §5.